EXECUTIVE REPORT

AI-Powered Telecom Customer Retention

A staff-approved workflow that turns churn alerts into timely, accountable customer care.

P — **Problem**

D — **Data**

M — **Method**

T — **Tools**

From prediction to approved customer action.

FPT Telecom · Internet services

Advanced Business Analytics

Group 3 · October 2026

Executive Overview

From churn prediction to approved customer action

FPT Telecom’s Internet business faces rising churn: national churn is up 20% year to date, and Binh Duong churn almost doubled from August to September 2026. **A churn model already flags high-risk customers, but a score alone does not tell staff why a customer may leave, who to contact first or what to do next.** This project builds a prototype decision system that turns each risk alert into an accountable care case: the model scores and explains risk, AI prepares drafts, staff review and approve every customer-facing action, and the system records what happened.

+20%
National Internet churn count, YTD 2026 vs. 2025

+92.3%
Binh Duong monthly churn, Aug → Sep 2026 (1,300 → 2,500)

0.9483
Validation ROC AUC of tuned LightGBM (synthetic data)

42.18bn
VND · illustrative Year-1 benefit, nationwide scenario

The PDMT framework at a glance

Each block answers one question and hands its answer to the next.

P Problem

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What decision is at stake?

- **Rising churn:** +20% nationally; Binh Duong 1,300 → 2,500 in one month
- **Alert-to-action gap:** staff see risk scores but unsure how to approach
- **Late contact and unresolved issues**

D Data

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What information does each decision need?

- **Customer data:** profile, plan, billing, service, usage, support history
- **Conversation data:** messages, replies, approvals, follow-ups, outcomes
- **ML & SHAP results:** churn scores and the drivers behind each score

M Method

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How is information turned into action?

- **ML churn prediction:** LightGBM vs. CatBoost → tuned LightGBM
- **SHAP explanation:** 127 features grouped into six business drivers
- **AI agents:** explain the case, draft messages and call scripts, draft tickets from chats

T Tools

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What delivers it?

- **Python + ML libraries** for scoring and SHAP
- **Supabase / PostgreSQL** and **React / Vite** for data and the staff workspace
- **n8n** with an **LLM + demo knowledge base** for AI agents

How the report flows

Problem p. 3 → **Data** p. 4 → **Method** p. 5 → **Tools** p. 6 → **Value & Risks** p. 7 → **Roadmap** p. 8

AI assists staff; it never contacts customers on its own. Data are public fictional and synthetic; delivery channels are simulated.

Problem

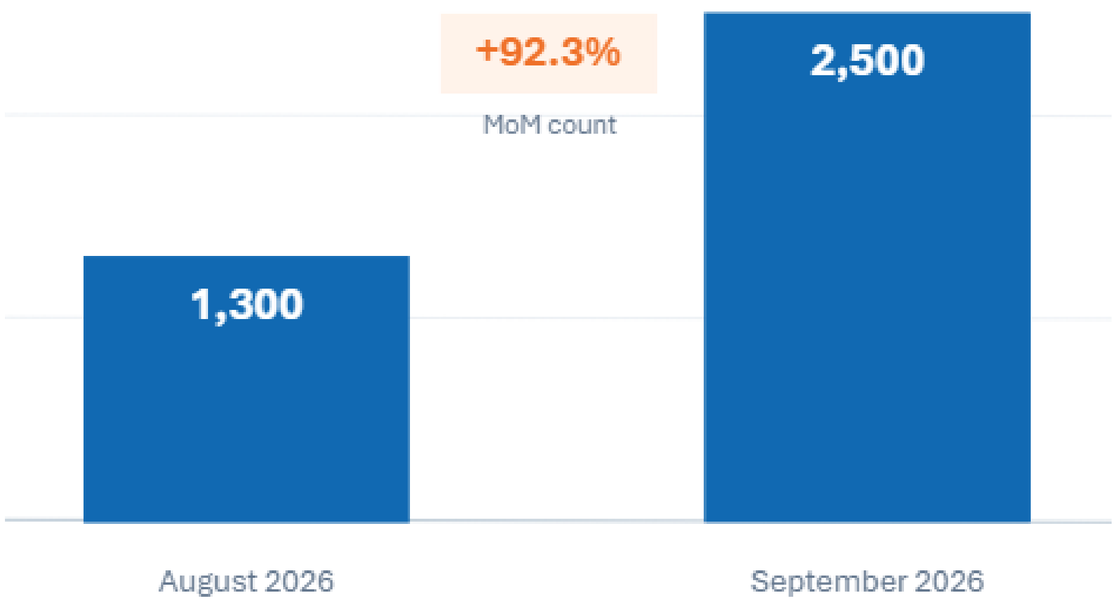
Why do churn alerts not turn into timely, effective customer care?

FPT Telecom

Internet services

Business context: churn is rising

Binh Duong · reported monthly customer churn



+20%

National Internet churn count, YTD 2026 vs. comparable 2025 period

1.19%

September churn rate, Binh Duong: 2,500 / ≈210,000 customers

Revenue exposure · same 2,500-customer cohort

625M	3.75B	7.5B
1 month	6 months	12 months

VND; ARPU VND 250,000/month assumed; no return, unchanged fee.

Higher churn increases the need for timely, accountable customer care.

Current process and operational gaps

Existing churn model → **Score >500 customer list** → **Monthly CS handover** → **Customer contact**

Reported gap	Operational difficulty	Decision needed
Unclear approach	Risk signal without enough context	**Why might this customer churn?**
Late contact	Some customers churn before contact	**Who needs attention first?**
Unresolved issue	Contact may not address the concern	**What action or handoff is needed?**

Who decides, and what they need

Stakeholder	Key decision	Information needed	Expected value
CS Agent	Choose approach and action	Customer facts · SHAP · history	Relevant customer care
CS Supervisor	Assign and monitor cases	Queue · ownership · progress	Clear accountability
Data Analytics	Evaluate prediction utility	Model version · score · outcome	Traceable feedback
Technical Support	Investigate service issues	Symptoms · support requests	Appropriate handoff

Agents and Supervisors are the direct users; Data Analytics and Technical Support provide evidence and escalation.

Project objectives

01 Prioritize and understand

Review churn alerts with customer facts and model explanations.

02 Prepare and coordinate

Use AI drafts, call scripts and support tickets to prepare the next action.

03 Preserve accountability

Track ownership, human approval, follow-up and period outcomes.

Next → Data: each decision above defines the information the system must hold: customer facts, conversations and model-explained risk.

D Data

What information does each decision need, and where does it come from?

Three data sources serve the three information needs

Data source	Available information	Role in the prototype	Serves
01 Kaggle dataset	Demographics, plan / contract, service flags, billing attributes, churn fields	Public fictional source; 7,043 customers	Customer data
02 Synthetic ML data	Generated service, billing and support histories; as-of snapshots; 127 model features	ML training, churn scoring and SHAP explanations	ML & SHAP results
03 Prototype / AI records	AI drafts, mock conversations, support tickets, approvals and follow-up	Simulated customer-care workflow	Conversation data

From public seed to one scoring batch

7,043 Kaggle customers 38 columns → **5,517** Internet cohort customer seeds → **113,937** As-of snapshots 127 features → **2,705** Customers in the 10 Oct scoring batch → **207** Flagged cases (score >500)

Different grains: source customers → Internet cohort → repeated snapshots → one scoring batch → cases for staff.

Time-aware snapshots

- **Three snapshots a month:** previous month-end, day 10 and day 20; predictions refresh on days 1, 11 and 21.
- **Only information available at the cutoff** is used; IDs and target labels are excluded.
- **Target:** churn within the next calendar-month window.



Supabase data model

28 tables

One shared case record links customer, ML, case, conversation, approval, ticket and outcome data, with role-based access.

- Customer & ML
- Cases & Messaging
- AI
- Support & Outcomes

Example customer snapshot record

CUSTOMER 0196-VULGZ · SNAPSHOT 10 OCT 2026 · PREDICTION WINDOW 11 OCT – 10 NOV 2026

01 Customer & subscription

Internet plan
Demo Internet 500 Mbps
Tenure
15 months
Monthly charge
VND 300,000
Billing mode
Postpaid

02 Usage & service

Usage (30 days)
284.00 GB
Disconnections (30d)
1
Downtime (30d)
15.26 min
Support contacts (30d)
0

03 Billing & payment

Fee change (30d)
0%
Current overdue
VND 600,000
Longest overdue delay
30 days
Contract
DEMO-0196-VULGZ

04 Churn prediction

Calibrated probability
60.67%
Risk score
606.74 / 1,000
Selected model
LightGBM tuned
Observed churn
Unknown

Synthetic demo record · not a real FPT customer.

Next → Method: these snapshots train the churn model; its scores and SHAP drivers then feed the AI-assisted care workflow.

M Method

How does a risk score become an explained, staff-approved action?

Literature-informed design: each limitation shapes one step

01 Explain risk

Model risk factors do not prove what caused churn.

Churn model + SHAP: 127 inputs grouped into 6 risk areas, shown as signals, not causes.

02 Create a case

A risk score assigns no owner and no next action.

Every alert becomes an assigned case in the care queue with customer details and next step.

03 Assist staff

AI suggestions may be incomplete or miss context.

AI drafts explanations, messages and call guides; staff check each one before contact.

04 Record action

Contact alone does not prove the customer was retained.

Review → send → reply; the case keeps messages, follow-ups, result and cost.

Grounded in four research streams: churn prediction & explainability, profit-driven retention, targeting by intervention effect, and AI assistance for service staff.

Model training and selection

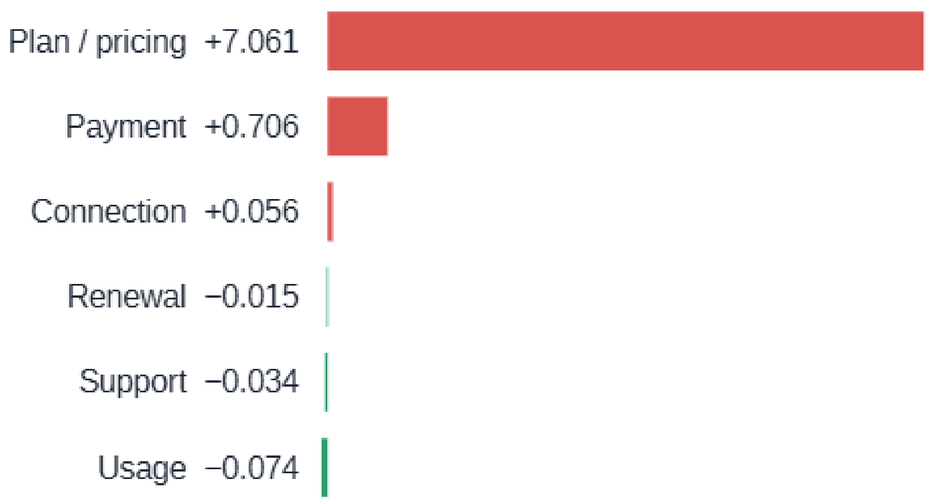
Candidate	Val. AUC	Test AUC	Test AP
LightGBM baseline	0.9469	0.9473	0.7166
LightGBM tuned ✓	**0.9483**	**0.9467**	**0.7138**
CatBoost baseline	0.9365	0.9374	0.7183
CatBoost tuned	0.9356	0.9348	0.7253

Selected by the highest validation ROC AUC under the predefined rule; test results are exploratory (synthetic data).

- **Gradient-boosted trees:** LightGBM and CatBoost capture non-linear effects in tabular data, on the same 127 features.
- **Temporal split:** January-start windows train, July validate, September test.
- **Optuna TPE:** 30 trials per family, three chronological folds, mean ROC AUC.
- **Leakage control:** purge unfinished target windows; hold out calibration and audit customers.
- **Platt calibration** adopted only if Brier score and log loss both improve; score = probability × 1,000, alert above 500.

Explaining each score: SHAP groups + AI explanation

Example: six signed SHAP groups (calibrated log-odds)



Predicted probability 96.96%

AI interpretation (for staff review): Plan and pricing is the strongest risk driver. The monthly fee rose 96.17% over 60 days to VND 563,000, and the customer requested three plan changes in 30 days. Suggested action: clarify the fee increase and discuss eligible alternative plans.

SHAP explains the prediction, not the confirmed cause; the AI never changes the score.

AI assistance and supervisor tools

AI Chat & Call

Customer facts, care context and the latest reply → evidence check, approved FAQ → chat draft, call script and staff guidance.

AI Chat-to-Ticket

Chat messages → issue summary, ticket fields and missing details → editable draft; staff decide when to submit.

Supervisor tools

Dashboard of workload, overdue follow-ups and outcomes; Manual, Round-robin or Weighted assignment, previewed before confirming.

Next → Tools: the method runs on five connected tools, with staff approval built into every customer-facing step.

T Tools

Which tools deliver the workflow, and does the prototype work?

Technology architecture: from prediction to action in one system

Python + ML
Scoring & SHAP (offline)

→

Supabase / PostgreSQL
Shared case state, sign-in, role access

→

React / Vite
Agent, Supervisor & customer portal

→

n8n
AI requests, mock delivery, case updates

→

LLM + demo knowledge base
Explanations, messages, call scripts

Tool	Role in the solution	Why we chose it
Python + ML libraries	Prepare features, train and compare models, predict churn, calculate SHAP values	Repeatable experiments and detailed model evaluation
Supabase / PostgreSQL	Store customer data and case records, manage sign-in, control access by role	Consistent data; access limited to authorized users
React / Vite	Agent and Supervisor interfaces and the simulated customer portal	Interactive case review and care management
n8n	Connect AI requests, simulated message delivery and case updates	Visual workflows that are easy to inspect and maintain
LLM + demo knowledge base	Use customer facts, SHAP results and approved guidance to draft explanations, messages and call scripts	Turns available information into readable drafts for staff review

Prototype demonstration **Web app:** ai-powered-telecom-customer-retention.vercel.app · video demo

Supervisor
Dashboard, case assignment and outcome monitoring

Agent
Case review, AI drafts, approvals and support tickets

Customer
Simulated portal and chat; no internal risk data shown

System test results

Three roles, three customer cases (active chat, no customer messages, failed care). Acceptance: accurate output, correct access and safe actions.

Test area	Case tested	Expected behavior	Observed result	Status
Access and privacy	Agent, Customer, Supervisor	Agent sees only assigned cases. Customer sees no internal data. Supervisor sees management tools.	Agent opened assigned case; unassigned case was not visible. Customer portal showed no scores, AI drafts or internal tickets. Supervisor had Dashboard and Assignment.	PASS
Explanation / message	No-customer-messages case (8443)	AI drafts for staff review. Nothing reaches the customer without staff approval.	Explanation shown to staff only. Message draft placed in the composer; not sent until staff clicked Send.	PASS
Call script	Active-chat, No-customer-messages, Failed-care (2905)	Scripts can be generated on open cases. New AI requests are blocked once a case is closed.	Scripts generated on open cases, including a staff-guided rewrite. New request blocked on failed case 2905.	PASS
Ticket and clarification	Active-chat case (8146, Supervisor)	AI drafts a ticket from chat evidence and lists missing details. Staff decide whether to ask the customer or submit the ticket.	Ticket drafted with quoted chat evidence; clarification message drafted. Neither was sent or submitted automatically.	PASS
Empty-chat protection	No-customer-messages case (8443)	AI ticket drafting is unavailable when there are no customer messages to cite.	“Draft ticket from chat” stayed disabled until customer messages existed; manual draft remained available.	PASS
Closed-case sending guard	Failed care (2905)	Send is disabled in the workspace once a case is closed (Failed or Care completed).	Send to simulated Zalo stayed disabled on the failed case, even with a non-empty draft.	PASS
Reply and care outcome	Active-chat (8146); Failed-care (2905)	Customer reply is saved and shown to staff. Failed care does not mean the customer left.	Reply persisted after reload and appeared in the Supervisor chat. Failed case kept retention outcome Unknown.	PASS
Supervisor dashboard	October cohort, one Agent	Filters and drill-down counts match. Outcomes stay unconfirmed until the period ends.	Agent had 51 October cases, 40 open; drill-down listed 40. All 51 outcomes showed “Awaiting period end”.	PASS
Assignment preview	Manual, Round-robin and Weighted	Supervisor previews and confirms assignments within capacity. Ownership, workload and Agent access update correctly.	Assignments respected capacity and persisted after reload. Workload, notifications and Agent access reflected the confirmed owner.	PASS

Next → Value & Risks: the prototype shows the workflow is safe and traceable; the next question is what it is worth and what could go wrong.

Business Value and Key Risks

What the workflow could be worth, and what must be controlled

Illustrative nationwide, full-coverage scenario (Year 1 of operation). Reuses the existing prediction model, SHAP and customer-care platform; not the Binh Duong pilot.

Benefit: additional customers retained

Assumption	Existing	Proposed
High-risk customers / month	90,000	100,000
Receiving timely care / month	50,000	**70,000**
Continued service, cared	60%	**62%**
Continued service, no timely care	30%	—

Customer group	Per month	Rate change	Retained
Already receiving timely care	50,000	60% → 62%	1,000
Newly receiving timely care	20,000	30% → 62%	6,400
Total	**70,000**		**7,400**

Value mechanism: AI-assisted preparation frees agent capacity, extending timely care from 50,000 to 70,000 customers a month. Most of the benefit comes from this added coverage.

7,400
additional retained customers a month · 88,800 a year

11.38%
relative reduction in baseline churn (≈780,000 churners a year nationally)

VND 100k
monthly contribution per retained customer (VND 250,000 ARPU × 40%)

VND 42.18bn
Year-1 benefit: 88,800 × VND 100,000 × 4.75 months

Read with care: 4.75 months is the average paid period recognised in Year 1 when retention is spread evenly (up to six extra paid months each). The assumed 62% continued service for newly cared customers (+32 pts) drives ~86% of the benefit and must be validated by A/B testing and a permanent holdout. Care contact alone does not prove retention.

Key risks and responses (likelihood × impact, scale 1–5)

ID	Risk	Description	L x I	Decision
R1	**Privacy, consent and channel rules breached**	Contacting opted-out customers, breaking Zalo rules, duplicate contact (predictions refresh on days 1, 11, 21), or chat data sent to an outside AI provider.	**4 x 5 = 20 Critical**	**Avoid**: recheck consent at approval and delivery; one owner per case; privacy sign-off before launch.
R2	**Unsupported AI content**	The AI may invent debt, discounts or reasons, or treat SHAP as proof. The prototype writes English only; its knowledge base is demo FAQ.	**2 x 3 = 6 Medium**	**Mitigate**: staff approve every message; approved offers and FAQ only; native-speaker review; pause if errors rise.
R3	**Unproven economics**	Break-even needs 65% of the base-case uplift (≈6.8 pts of extra retention on 70,000 cared customers). The uplift is assumed, and high-risk customers are not always persuadable	**4 x 4 = 16 High**	**Accept (capped)**: fixed pilot budget; validate uplift by A/B test and holdout; no-go if uplift is below break-even.
R4	**Staff capacity and adoption**	Staff review every AI draft, and may skip a tool that is hard to use.	**3 x 3 = 9 Medium**	**Mitigate**: cap the list to capacity; rank overdue cases first; training and usability test.
R5	**Policy and ownership decisions pending**	Offer catalog, follow-up delay, CS sponsor and owners are not set. A technical handoff proves a request was sent, not that the fault was fixed.	**4 x 3 = 12 High**	**Mitigate**: Stage 2 gate with named owners; Finance-approved offer catalog; track technical handoffs.
R6	**Access-control or security failure**	22 permission-bypassing functions are flagged, leaked-password protection is off and concurrent use is unproven. Not a production sign-off.	**2 x 4 = 8 Medium**	**Mitigate**: review flagged functions; enable password protection; run access and double-approval tests.

Next → Roadmap: because economics (R3) and privacy (R1) are decisive, the rollout tests uplift in gated pilots with a capped budget before any scale-up.

Implementation Roadmap and Next Steps

Phased rollout with go / revise / stop gates

STAGE 1 · CORE PRODUCT ENGINEERING

Core components, demonstrated in the prototype

01 Data & features

Rolling-month windows; 127 metrics; purged splits

02 Tune LightGBM

Val. AUC 0.9483; Platt-calibrated; alert >500

03 SHAP risk

Native TreeSHAP; 6 approved groups; top-3 drivers

04 Care queue & AI

Priority queue; AI explanation; grounded facts

05 Outreach review

Zalo & call drafts; eligibility checks; mandatory approval

06 Chat-to-Ticket

AI ticket extraction; follow-up reminders

07 Prototype QA

73 DB & UI checks passed; Agent vs. Supervisor access

PRE-PILOT GATE
Enterprise data & policy sign-off

STAGE 2 · PHASED PILOT IN BINH DUONG

Feature-by-feature rollout to 60 agents, each followed by a gate review

P1 Diagnostic AI

- Roll out Care Queue & SHAP
- Read-only AI explanation before regular calls
- Staff validate risk drivers

GATE 1
- Pre-call review time cut by >30%
- AI factual accuracy >90%

P2 Assisted outreach

- AI drafts personalised Zalo / call scripts
- Mandatory staff review and approval
- Controlled A/B pilot vs. standard care

GATE 2
- Draft acceptance / edit rate >70%
- Reply rate rises; opt-out <2%

P3 Chat-to-Ticket

- AI turns complaints into tech tickets
- Follow-up reminders
- Track period-end retention

GATE 3
- 0 tickets returned for missing evidence
- Resolved cases churn less

Gate review GO → next feature, then Stage 3 · NO-GO → revise or stop

STAGE 3 · ENTERPRISE EXPANSION & SCALING

Controlled multi-province expansion with break-even governance

W1 3–5 provinces

Mix of high and moderate churn; 2–3 month holdout; check SHAP drivers and acceptance replicate

W2 10–15 provinces

Stress-test concurrency and live scoring; audit cost vs. retained margin; LLM and n8n budgets

W3 Nationwide

All care centres; core CRM and ticketing integration; 5% permanent holdout for ROI audit

C Continuous learning

Monthly outcome feedback; active vs. churned ground truth; retrain LightGBM for drift

Conclusion: what the PDMT analysis shows

P The real gap is operational: churn alerts were not becoming timely, accountable action.

D Customer, conversation and model data now meet in one 28-table case record.

M Tuned LightGBM (validation AUC 0.9483, synthetic) and six SHAP groups prioritise and explain; AI drafts, staff decide.

T A working prototype on five connected tools passed all seven system test areas across three roles.

Next steps Secure enterprise data and policy sign-off → run the Binh Duong pilot feature by feature with go / revise / stop gates → scale only where measured uplift beats break-even.
From prediction to approved customer action.